# Projecting Computing History

## A Hybrid Live-Virtual Visit to The National Museum of Computing

Claire Marston
Head of Learning
The National Museum of Computing
Milton Keynes, UK
claire.marston@tnmoc.org

James Wolfer
Department of Computer Science
Indiana University South Bend
South Bend, IN, USA
jwolfer@iusb.edu

***Abstract*—History and travel are valuable components for building cultural awareness and preparing students to work in a global economy. In many STEM disciplines history is an intrinsic component of the pedagogy, serving to bind students to the professional community and culture. Computing, in contrast, has not systematically embedded historical awareness into the curriculum. As a step toward exposing students to computing history and enhancing cultural awareness, this work profiles a collaborative effort to take classes on a virtual field trip to The National Museum of Computing in the UK, with live faculty interaction across international borders.**

***Keywords—pedagogy; computing history; virtual museum***

## I. Introduction

History and travel are valuable components for building cultural awareness and preparing students to work in a global economy. Many STEM (Science, Technology, Engineering, and Mathematics) disciplines, such as physics, weave history into the curriculum. They also embed history into the discipline itself, honoring historic figures by assigning their names to units of measurement, e.g. Hz, Ohm, and Pa. This is in contrast to bit, byte, etc. in computing. John Impagliazzo summarizes the situation as "Unlike some other fields of study, computing professionals tend not to promote and preserve their own history in a broad and consistent manner" after asserting that "History has a way of encouraging students to look beyond the machine and broadens their outlook on the social implications of computing" [1,2]. Likewise, Lee sees computing history as a vehicle for integrating computer concepts into the general education curriculum for all students: "So far the history of computing, along with computer ethics, has been one of the vehicles by which we have been able to integrate these into the CS curriculum, and as far as can been seen may continue to be the medium for future initiatives" [3].

As a step toward exposing students to computing history and enhancing cultural awareness, this work profiles a collaborative effort to take classes on a hybrid live-virtual field trip to The National Museum of Computing based in the UK, with live faculty interaction across international borders. Specifically, through close collaboration between the university and the museum, nearly one hundred and thirty students participated in a remote teaching exercise with custom-curated presentations on the history of computing. The balance of this work describes the participating classes, collaboration, and presentations developed in the course of this virtual field trip.

## II. Virtual Field Trip

The concept of a virtual experiences in engineering and computing education is well established. From virtual laboratories to virtual reality and computer games, many formats are designed to provide remote experiences without necessarily duplicating physical environments [4]. Likewise, field trips have long been a part of higher education. One interesting paper, Shakil et.al describes a comprehensive survey of over 150 university students and faculty to assess the value of field trips in general. Their conclusion was that field trips promote leadership, socialization, and discipline among students. They also show that the majority of students believe that field trips provide experience and observation opportunities, as well as promoting interaction between faculty and students [5]. Similar observations asserting positive results from engineering field trips have been reported by others [6,7].

While there has been a long history of educational field trips at almost every educational level, the concept of a virtual field trip is relatively recent. Tuthill describes the use of the virtual field trip in secondary education [8], including both positive and negative aspects of the experience. These include the positive attributes of scheduling ease, flexibility, and content control against the potentially negative aspects such as student comfort with the presentation technology, technology access, and associated equipment costs.

More recently, LoPiccolo compares the perceived difference in value between real and virtual field trips in the context of the sustainable building. Pre- and post-trip testing for the real and virtual field trip students, each in a Materials and Methods of Construction course, showed that real field trips are more effective as a learning tool. In addition a perceptual survey indicated that 28% more of the actual field trip participants strongly agreed that they learned more about the topic on their trip. While the actual field trip was more effective, "the data also suggests that virtual field trips should be considered a dynamic, useful learning tool." [9].

Assuming that real field trips have an advantage over virtual field trips, and given the constrains of institutions separated by an ocean, we elected to try a hybrid approach in this work. In conjunction with other professional travel, and in

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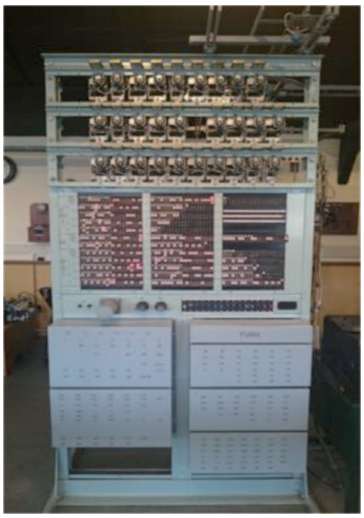

Fig. 1. Tunny

collaboration with The National Museum of Computing, the course instructor traveled to Bletchley Park and taught classes remotely, live, from the UK using a series of experiences custom curated for the classes involved. This provided an interactive experience for the students and serves to mitigate some of the static element of a totally virtual visit. In addition, a follow-up live remote visit by museum staff accentuated the experience by answering lingering questions raised by students. The balance of this work describes the classes, the curated experiences, and the remote teaching experience.

## III. Classes Involved

The university campus involved in this work is an urban campus consisting of approximately 7500 students offering over 100 undergraduate degrees as well as selected graduate programs. As an urban satellite campus of a large state university, this campus draws 95% of its students from the local community. Many of these students have not had the opportunity to venture beyond the local community and surrounding states, so any international exposure is a welcome addition to their educational experience.

The classes involved with this project are offered by the Computer Science and Information Systems department in the university. Specifically, two classes were taken on the virtual field trip. The first, a freshman level, general education class offered to non-computing majors, involved approximately 110 students in two sections. This class is designed to introduce students to the basic concepts, applications, and implications of modern computing. Topics include hardware, software, and security, as well as exploring ethical and societal aspects. In addition, basic skills necessary for college work such as word processing, spreadsheets, and database applications are taught.

The second class was a Parallel and Distributed Processing course consisting of approximately twenty students divided almost equally between senior undergraduates and graduate students. The class content includes a combination of high-performance parallel architecture, multiple parallel programming paradigms suitable for heterogeneous computing, and significant group projects.

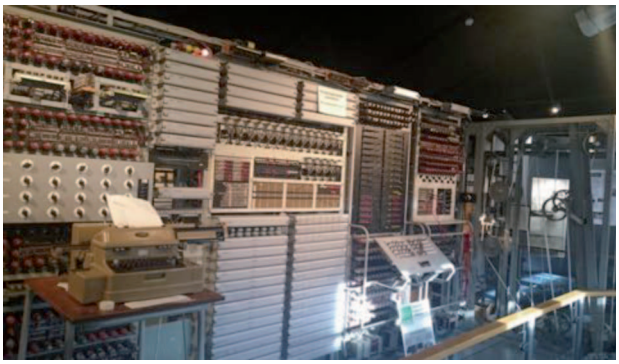

Fig. 2. Colossus

## IV. The National Museum of Computing

Bletchley Park, roughly an hour north of London, is the historic site of Second World War code breaking efforts. Among the elite occupants of the Bletchley Park campus was Alan Turing, often considered the father of modern computing for his published insights into the theoretical fundamentals of computing.

Located on the Bletchley Park site is The National Museum of Computing. This facility contains an amazing variety of computing artifacts, spanning from what is arguably the first programmable computer, the Colossus, to the foundations of modern computing. Working with the Head of Learning, we developed a custom set of experiences for the students. The constraints mandating a custom approach included:

1. Funding. This project was accomplished with volunteer participation, limiting the ability to have facilities open past normal operating hours.
2. Staff scheduling. There were limitations on staff availability such as holiday leave.
3. Time zone differences. There was a five hour time difference between the class location and Bletchley Park. Most of the classroom meetings occurred outside museum working hours.

To address these constraints while being consistent with our hybrid approach we elected to develop a series of video presentations prior to travel. These were archived at the classroom site so they would be available to show. In all, eleven video presentations were produced. In the spirit of the BBC History of the World in 100 Objects [10], we explore the history of modern computing through samples of machines located at the museum. Beginning with an introduction to the museum the presentation then turns to a description of the role of Alan Turing at Bletchley Park. While the Computer Science students are well aware of Turing's contribution to the theoretical principles of computing and his famous test for machine intelligence, many of the students in the computer literacy class have never heard of Turing before. Both groups are unlikely to grasp the enormous influence Turing had on the signals decryption community during the war, something the presentation directly addresses [11]. Subsequent presentations include [12-14]:


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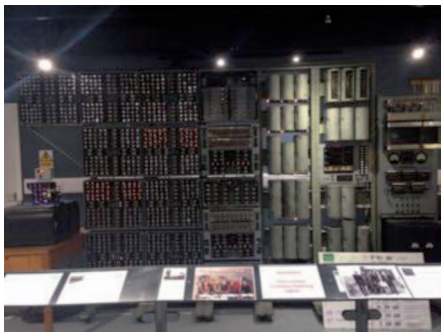

Fig. 3. Harwell Dekatron, aka WITCH

Fig. 4. Cray 1

### A. *The Tunny Gallery*

Designed by Bill Tutte, Tunny (Fig. 1) was designed to replicate the German Lorenz encryption device. It is amazing that this was accomplished through a combination of luck and intense reverse engineering without ever seeing an actual Lorenz machine; not least because the Lorenz machine is believed to have more settings than there are atoms in the known universe! None of the Allies saw a Lorenz machine until USSR soldiers captured one after the Battle for Berlin, at the end of the War in Europe. Without being able to access the original machine, or 'the book', which was used to set the wheel settings, teams of mathematicians based at Bletchley Park had to use mathematical calculations to work out both how the Lorenz worked, and the settings for specific messages. Tunny was incredibly significant to the Allies war effort, as the Lorenz machine was used by Hitler, and the German High Command to send large documents and orders. Tunny was how the Allies were able to decrypt these intercepted messages. An initial attempt to develop a machine to statistically search for Tunny settings, the Heath-Robinson, resulted in a significant speedup but suffered reliability issues with it paper-tape store. This is why Colossus was built.

### B. *Colossus*

Colossus, shown in Fig. 2, is arguably the world's first programmable, electronic computer. Built, like Tunny and Heath-Robinson, to decrypt messages during the Second World War, it was kept as a state secret until the 1990's [15]. Using paper tape for memory, the machine was used to calculate the settings for Tunny, in order to duplicate the Lorenz machine wheels, and therefore decrypt the message. Colossus contained 2400 thermionic valves (aka "tubes"), and could read the characters encoded on paper tape at the rate of 5000 characters per second and perform arithmetic and logic operations. Unlike Tunny, which was hardware-limited to a single operation, Colossus was designed to allow reconfiguration, i.e. programming, earlier than the well-known ENIAC. The Colossus housed at The National Museum of Computing is the only example in the world, of the 10 which were originally built in 1944. It is a rebuild, rather than an original, because whilst it contains some original parts, at the end of the Second World War all the machines were destroyed, so it is not possible to attribute parts to a particular Colossi (ie, it could be from Colossus 9 or 10).

### C. *Harwell-Dekatron, The WITCH*

The WITCH (originally called the Harwell Decatron), shown in Fig. 3, is the world's oldest original working computer. It has been recognized as such in the Guinness Book of World Records since 1973. It has 1/3 KB of memory, and was used to help build the UK's first nuclear power station (at Harwell in Oxfordshire) in 1950. The computer used a combination of 720 Dekatron tubes along with sequencing relays to do decimal calculations. The computer was first operated in 1950, and was in use for educational purposed until 1973, following the successful build of the Harwell Nuclear Power Station. The Harwell Decatron was given away in a competition to an educational establishment (Wolverhampton and Staffordshire College), who renamed it the Wolverhampton Instrument for Teaching Computation from Harwell (WITCH). It was ultimately moved to The National Museum of Computing where it still runs today, making it the world's oldest original working computer.

### D. *Elliott 803 and 903*

Working 1960s computers can be used to demonstrate how quickly computing developed. In only five years the size of the computer, in the Elliott 803, decreases by approximately 75% with the Elliott 903 These computers were commercially available, used germanium and silicon transistors respectively as their logic basis, and could optionally use magnetic tape as storage. The Elliott 803 is also a vehicle for introducing the significant role women had on early computing. For example, an Elliott 803 was used by Dina St Johnston to establish and run a computer service bureau. The computer supporting the enterprise was worth 10 family homes when new! Dina did all the programming herself, earning a huge amount of money per hour in the 1960s.

### E. *Early PCs*

Commodore PET and 64, DEC PDP 8, Apple Lisa and BBC Micros, are all used to demonstrate what personal and home computing looked like in the 1980s. These computers are especially interesting as they often trigger reminiscence. This is especially true since many students recall playing games on these machines or their close contemporaries.


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Furthermore, Computer Sciences students are exposed to a variety of CPU instruction sets, with illustrations from the CPUs of the 1980s forming a thread for discussing the computers they currently use on a day-to-day basis. Fundamental hardware and software paradigms, such as the computer mouse, windowing systems, and the UNIX operating system emerged within this timeframe.

### F. *Early Portable Computers*

Notably, the museum owns an Osbourne 1, which is regarded by some as the first laptop. Looking suspiciously like a portable sewing machine, the Osborne 1 weighs approximately 11kg, has a 5-inch screen, and needed a car battery to run, which also had to be carried. While not particularly practical by today's standards, the Osborne 1 set the stage for groundbreaking portables such as the Apple Macintosh portable – a true "laptop" computer.

### G. *Supercomputers*

Since one of the classes in session during the course of this remote field trip was Parallel and Distributed Programming, the presentations featured a special segment on the high performance artifacts in the museum. Among these artifacts include the iconic Cray supercomputers (Fig. 4), along with a running Transputer demonstration. The Transputer, designed by INMOS, was an early integrated circuit CPU designed for interconnected communication, anticipating the parallel programming paradigms in contemporary computing. For this demonstration, a video showing the acceleration for computing and displaying the Mandelbrot set as additional Transputer nodes are added.

## V. Remote Class Presentation

The actual classes were taught from the instructor's hotel room in nearby Milton Keynes. Budget constraints, staff holiday, and time zone constrained the delivery. For example, the local times for the three classes were 3pm, 7:30pm, and 9pm. The last class ended at 10:15pm local time. Since the museum was closed at night this necessitated actually teaching from an alternate location, thus the hotel room.

The class content consisted of both an expansion on the pre-recorded video presentations from the museum and an interactive discussion with students. In addition to the museum topics, a brief introduction to Bletchley Park and an expanded description of the role of Alan Turing in both codebreaking and computing. This was followed by a virtual tour of The National Museum of Computing through a combination of live instructor commentary and the pre-recorded video presentations. Student questions were entertained live, and normal classroom administrative activities, such as homework questions, were dispatched.

To facilitate both the presentations and the interaction at the classroom both a multimedia technician and a faculty facilitator to introduce the technician to the class and to ensure the class gets off to a good start. The faculty member remains available nearby for the duration of the event should help become necessary.

## VI. Observations

For the most part technical difficulties were minor. Initially there was concern over the potential of inadequate hotel internet speed. It was necessary to have a video connection to facilitate the teaching activities. Fortunately, the hotel had sufficient internet speed and the video conference went well.

One significant difficulty was the lack of audio consistency between the conferencing audio and the audio levels in the pre-recorded video. This required constant readjustment by the multimedia technician. In addition, the video segments were acquired over the course of several weeks, and the audio interface equipment was changed in the middle of the recording sessions. The audio environment also changed from museum gallery to gallery depending on local acoustics and currently operating machinery. An attempt to post-process the audio helped some, but was not completely successful in normalizing the audio.

It was not possible to acquire formal student feedback for this project as the class is still in session as this is being written, but there was a variety of informal feedback. Specifically, both the faculty support and the technical staff observed a considerable level of engagement on the part of the students. Several students indicated that they really liked the remote engagement, as well as the museum tour itself. The most significant questions came, as expected, from the Computer Science students taking the high-performance, parallel programming course. They especially expressed that the historic connections were illuminating and helpful.

## VII. Future Work and Conclusion

This collaboration between our university and The National Museum of Computing worked well, and the results were better than we had hoped for. Through considerable effort on behalf of the museum staff, the university staff, and faculty engagement, students who rarely, if ever, travel outside their local community were exposed to world-class exhibits directly related to their classwork.

As a result of this effort we are exploring a variety of avenues for future collaboration. Of course, repeating the tour and including a formal assessment is one important example. Other examples under consideration include extending this prototype for virtual visits to include additional class-specific exhibits. For example, Artificial Intelligence, Computer Graphics, or Game Design are all candidates for specific virtual tours. We are also exploring taking VR photos and tours, for use with 3D smart phones, such as those with Google Cardboard, with school groups. If funding can be acquired the possibility of teaching live from the museum across many time zones is a possibility. Finally, there may be opportunities for remote interactive activities, such as Raspberry PI, or Arduino projects.

In conclusion, this virtual visit to The National Museum of Computing exposed students to the history of computing, engaged them across international borders, and opens the potential for ongoing exploration between the participating institutions.


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